Literature Status: Banach Spaces Without Smooth K-Maps

FA/Banach run `fa_banach_001`

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Status

This is a literature-already-answered status packet, not an original proof from the run. Belitskii–Rayskin, arXiv:1703.06299, ask in Section 5, Open questions, item 1, whether Banach spaces without C^∞ K-maps exist. The existence subquestion has an affirmative later-literature answer.

Original Problem

The original source is:

> G. Belitskii and V. Rayskin, *Extension of local smooth maps of Banach spaces*, arXiv:1703.06299.

The paper defines a C^q-K-map on a Banach space X as a globally defined bounded C^q representative of the identity germ at the origin. Equivalently, it is a bounded C^q map $H : X \to X$ which agrees with the identity on a neighborhood of 0. Its open-questions section asks: for which spaces do C^∞-K-maps exist, and do Banach spaces without C^∞-K-maps exist?

Later Answer

The supporting source is:

> G. Belitskii and V. Rayskin, *A New Method of Extension of Local Maps of Banach Spaces. Applications and Examples*, arXiv:1706.08513.

This later paper uses the name *blid map*, defined as a globally bounded local identity. Thus the notion is the same as the earlier K-map notion. In its section "More examples and open questions", it says that the question whether Banach spaces without smooth blid maps exist has an affirmative answer, citing D'Alessandro–Hajek and Hajek–Johanis. The stated reason is that those works provide Banach spaces which do not allow C^2-extension, and hence cannot have a C^2-blid map. Therefore these spaces cannot have a C^∞-K-map either.

Supporting Bibliography

- S. D'Alessandro and P. Hajek, *Polynomial algebras and smooth functions in Banach spaces*, Journal of Functional Analysis **266** (2014), 1627–1646, DOI 10.1016/j.jfa.2013.11.017.
- P. Hajek and M. Johanis, *Smooth Analysis in Banach Spaces*, de Gruyter, 2014.

Verification Notes

The answer is not contained in arXiv:1703.06299; there the question appears as an open question. The later arXiv:1706.08513 source is distinct and explicitly records the affirmative answer to the same existence question under the equivalent "blid map" terminology. The implication from no C^2-blid map to no C^∞-K-map is immediate, since every C^∞ K-map is a C^2 bounded local identity.

Search Bounds

The local run registry, ledger, solution folders, and attempts were searched for arXiv:1703.06299, arXiv:1706.08513, K-map, smooth K-map, blid map, bounded local identity, and the non-even ℓ_p subquestion. Existing nearby records cover the harder non-even ℓ_p blid question and a weak-topology partial result, but no packet for this exact existence-status subquestion was found. The later arXiv:1706.08513 source and bibliography were inspected, and the D'Alessandro–Hajek DOI was checked by web search.

Limitations

This packet records only the affirmative answer to the existence of Banach spaces without smooth K-maps. It does not classify spaces admitting K/blid maps. It does not answer the non-even ℓ_p question, the sphere question in $C[0,1]$, the arbitrary-subspace question for $C[0,1]$, or the infinite-dimensional Whitney-extension question.

Human Review Recommendation

Check the terminology identification between K-maps and blid maps, and decide whether to attach the underlying D'Alessandro–Hajek JFA PDF if publisher access permits. The packet's scope is intentionally narrow: existence of spaces without smooth K-maps is already answered; the surrounding classification questions remain open here.